

FUNCTIONAL SPECIFICATION DOCUMENT

Car Rental and Equipment Management System (CREMS)

Motors • Carptrac • Limited Shipping Hire Services

Purpose: Defines exactly how CREMS screens, workflows, validations, permissions, calculations, documents and integrations shall operate for implementation and client acceptance.

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Prepared for	Carpenters Fiji Pte Limited
Prepared by	CS400 IEP Project Team
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Initial functional scope	Carpenters Motors; Carptrac; Carpenters Shipping limited to portable toilets, large bins and scaffolding hire
Related baseline	CREMS Software Requirements Specification v1.0 and validated client meeting decisions

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Document control

Version	Date	Author / owner	Change	Approval
0.1	19 Aug 2026	CS400 IEP Project Team	Initial FSD derived from requirements and implemented CREMS baseline	Internal review
1.0	19 Aug 2026	CS400 IEP Project Team	Issued for Carpenters functional validation	Pending client sign-off

Review and sign-off

Reviewer	Role	Decision	Date	Signature / comments
	Carpenters business representative	Approve / Approve with changes / Reject		
	Carpenters IT / security representative	Approve / Approve with changes / Reject		
	Project supervisor	Reviewed		

Sign-off meaning: Approval confirms that the described functional behaviour is an agreed delivery baseline. Later changes are controlled through the change-request process; sign-off does not waive security, data-quality or acceptance-test obligations.

Distribution

Controlled copies: Carpenters project stakeholders, CS400 project team and academic supervisor. Credentials, production secrets and personal customer data are excluded from this document.

Contents

1. Introduction and boundaries	
2. System context and functional architecture	
3. Roles, permissions and operating scope	
4. Common user-interface behaviour	
5. Authentication, sessions and access administration	
6. Organization, divisions, branches and services	
7. Customer and customer-portal functions	
8. Asset register, QR, lifecycle and availability	
9. Booking requests, quotations and approvals	
10. Pickup, agreements, active hire and returns	
11. Inspections and maintenance	
12. Personnel, transport and field operations	
13. Finance, profitability, dashboards and reports	
14. Notifications, documents and scheduled processing	
15. API, validation, errors and audit behaviour	
16. Functional acceptance and traceability	
17. Assumptions, open decisions and change control	
18. Client sign-off	
Appendix A. Status models	
Appendix B. Output document catalogue	
Appendix C. Functional requirement index	

1. Introduction and boundaries

1.1 Purpose

This FSD translates the approved CREMS requirements into observable system behaviour. It is intended for client validation, UI and API implementation, database design, test-case preparation, user training and acceptance testing. “Shall” statements are mandatory for the signed baseline.

1.2 In-scope divisions and services

Division / area	Assets and services	Division-specific behaviour
Carpenters Motors	Passenger and commercial rental vehicles	Registration, odometer, fuel, driver/licence checks, vehicle inspections and daily/weekly rates.
Carptrac	Forklifts, cranes, excavators, generators and related hire equipment	Engine hours, capacity/specification fields, operator qualification and assignment, safety checks, hourly/daily rates and delivery where required.
Carpenters Shipping – limited hire scope	Portable toilets (portaloos), large metal bins and scaffolding	Quantity or unit-based hire, delivery/collection, site instructions, condition/cleaning checks and transport/labour/setup charges. Freight, cargo, vessel and port operations are excluded.

1.3 Functional boundaries

- Public catalogue and availability search; authenticated customer booking and tracking.
- Staff role- and permission-based portals scoped by division and branch.
- Customer, asset, quotation, booking, approval, agreement, inspection, return, maintenance, operator, finance and reporting workflows.
- Flexible service catalogue, category attributes, rate and charge definitions so future divisions can be configured with minimal code change.
- QR-based asset identification, email/PDF outputs, scheduled alerts and immutable audit events.

1.4 Out of scope for this baseline

- Shipping freight forwarding, vessel, cargo, port and container-terminal operations.
- Live online payment-gateway settlement, GPS route optimization and offline mobile synchronization unless approved through change control.
- Replacement of Carpenters’ accounting, payroll or HR systems; CREMS will remain integration-ready.
- Production deployment, data migration and third-party commercial fees unless separately authorized.

2. System context and functional architecture

Layer / actor	Responsibility	Information exchanged
Public and customer web	Browse services, register/sign in, request a booking, track status, view quotations, agreements and invoices.	HTTPS JSON requests; authenticated customer session; downloadable documents.
Staff web portal	Role-specific work queues for bookings, assets, maintenance, finance, reporting and administration.	HTTPS JSON requests; secure staff session; QR input; document actions.
ASP.NET Core API	Validates identity, scope and input; applies business rules; coordinates persistence, documents, email and audit.	REST endpoints under /api; consistent status codes and correlation identifiers.
SQL Server	Stores operational records, configuration, security history and audit data.	Transactional reads/writes through Entity Framework Core.
Email service	Delivers verification, reset, quotation, confirmation, agreement, invoice and alert messages.	Queued message, approved recipient, attachments/links and delivery result.
PDF / QR services	Generate immutable customer documents and asset identification labels.	Versioned document snapshots and non-sensitive QR token/code.

2.1 Functional design principles

- Configuration before customization: divisions, branches, services, categories, attributes, charges and approvals are data-driven.
- Least privilege: every protected action checks account state, role/permission and division/branch scope on the server.
- Guided work: one obvious primary action, visible prerequisites and staff-friendly status labels.
- Traceability: critical actions create lifecycle and audit records with actor, timestamp, scope and reference.
- Performance: list screens use filtered queries, server pagination and selective fields rather than loading full histories.

3. Roles, permissions and operating scope

Role	Functional responsibility	Default operating scope
Super Administrator	Group configuration, divisions, permissions, security and full oversight.	All divisions and branches.
Administrator	Staff/customer access administration, approved configuration and audit.	Assigned or group-wide; cannot alter a Super Administrator unless explicitly permitted.
Branch Manager	Branch operations, approvals, exceptions and branch financial performance.	Assigned division(s) and branch(es).
Rental Officer	Customer records, requests, quotations, bookings, pickup and return.	Assigned division/service and branch scope.
Maintenance Officer	Faults, maintenance jobs, parts, meters and return-to-service checks.	Assigned maintenance-capable branches/divisions.
Finance Officer	Invoices, allocations, credit notes, statements and financial reports.	Assigned operating scope.
Driver / Operator	Assigned deliveries/hire work, QR scans, meters, checks and timesheets.	Only assigned jobs/assets and permitted field actions.
Customer	Own profile, bookings, quotations, agreements, invoices and requests.	Own account and authorized corporate account only.

3.1 Permission rules

Permission family	Examples	Rule
Identity	users.create, users.reset_password, users.manage_access, customers.manage_access	Server evaluates permission and target-user protections; self-elevation is blocked.
Organization	divisions.configure, branches.configure, services.configure, branch_calendar.manage	Division creation/deactivation is reserved for Super Administrators.
Assets	assets.view/create/edit/transfer/inspect/record_meter/retire/view_financials	Asset branch/division scope is checked on every read and mutation.
Operations	rentals.approve, maintenance.complete, pricing.configure	Approval separation and value/workflow rules apply.
Finance	payments.refund, reports.financial	Financial details remain hidden without explicit permission.

Scope precedence: An assigned role or permission never grants access outside the user's active division/branch scopes. Temporary access must have start and end dates and is ignored after expiry.

4. Common user-interface behaviour

UI element	Required behaviour	Validation / accessibility
Navigation	Public site and customer portal are separate from /staff. Staff menu is grouped as Work, Fleet, Insights and Administration and collapses on desktop.	Only accessible pages are rendered; direct navigation is also rejected by the API.
Breadcrumbs	Display Home and current workspace; selecting an ancestor returns without losing saved data.	Current page has aria-current; browser back remains usable.
Work queues	Use status tabs with visible counts, server search, filters and pagination.	Empty, loading and error states explain the next action.
Record workspace	Use a full page or side drawer with summary, requirements, actions and collapsed history.	One primary action; destructive or irreversible actions require confirmation.
Forms	Labels remain visible; required fields and examples are explicit; dependent fields appear only when relevant.	Inline validation plus summary; focus moves to first invalid field.
Notifications	Success confirms record/reference and next action; warning explains recoverable issue; error includes safe retry guidance.	No credentials, stack traces or personal data in UI errors.
Responsive layout	Desktop tables adapt to mobile cards; key action remains visible.	Keyboard navigation, contrast, semantic headings and labelled controls are required.

4.1 Standard list behaviour

- Default page size 25; permitted sizes 25, 50 and 100.
- Filters persist during the current workspace session and can be cleared in one action.
- Sorting occurs on the server for indexed fields such as reference, status, date, branch and asset number.
- Financial columns and actions are omitted when the actor lacks permission; values are not merely hidden with CSS.

5. Authentication, sessions and access administration

Provide separate secure authentication experiences for customers and staff while enforcing account lifecycle, session timeout and least privilege.

Specification item	Definition
Primary actors	Staff, customers, Administrators and Super Administrators
Required permissions / scope	Public login endpoints; protected account and access actions use explicit permissions.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

5.1 Staff authentication and session flow

ID	User action / input	System processing and rules	Result
F-AUTH-01	Enter staff email and password.	Normalize email; rate-limit attempts; verify password hash and active staff role; never reveal whether an email exists.	Authenticated session or generic failure.
F-AUTH-02	Submit MFA code when policy requires.	Verify active challenge, expiry and attempt limit. Super Administrator MFA may be disabled only as a documented temporary environment setting.	Session upgraded or challenge rejected.
F-AUTH-03	Open another browser window.	Window-bound token must match the server session. A copied cookie without the matching window token is rejected.	New window must authenticate separately.
F-AUTH-04	Remain inactive.	Expire idle session at configured timeout; absolute lifetime also applies; warning is shown shortly before expiry.	Protected calls return 401 and UI returns to staff sign-in.
F-AUTH-05	Log out or revoke a session.	Invalidate server session and cookie; record logout/revocation event.	Session cannot be reused.

5.2 Account and access administration

ID	Function	Key rules / validations	Audit outcome
F-IAM-01	Create/invite staff	Unique normalized email; role, division/branch scope and lifecycle state required. Temporary password must be changed.	Actor, target, role and scopes.
F-IAM-02	Assign scopes and permissions	Support one/multiple divisions/branches, temporary access and group-wide scope. Prevent self-elevation and invalid branch-division pairs.	Before/after permission and scope snapshot.
F-IAM-03	Reset password / unlock	Require permission; send reset workflow or issue temporary credential according to policy; revoke existing sessions.	Security event and notification delivery.
F-IAM-04	Lifecycle and departure	Invited → Activation pending → Active → Locked/Suspended/Deactivated. Departure revokes sessions, removes temporary access and approval assignments.	State transition and affected assignments.
F-IAM-05	Security history	Display login success/failure, IP, approximate device, MFA, reset, logout and expiry. Allow own sessions to be revoked.	Read access itself is logged where required.
F-IAM-06	Separation of duties	Block self-approval, self-role elevation, normal-admin changes to Super Administrators and creator approval of sensitive refunds.	Denied action with reason code.

6. Organization, divisions, branches and services

Configure the group hierarchy and rental capabilities without requiring code changes for every future division.

Specification item	Definition
Primary actors	Super Administrator; Administrator; authorized Branch Manager
Required permissions / scope	divisions.configure, branches.configure, services.configure, branch_calendar.manage, pricing.configure
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

ID	Function	Inputs and validation	Functional result
F-ORG-01	Maintain division profile	Unique code/name; branding, contacts, currency, tax, terms, capabilities, public/active flags and default workflow.	Division becomes selectable only when active.
F-ORG-02	Maintain service catalogue	Division, type, hire unit, personnel/delivery/quote rules, required documents, deposit, meter and inspection configuration.	Service can be enabled per branch.
F-ORG-03	Maintain branch profile	Code/name, addresses, contacts, coordinates, manager, instructions, delivery coverage and public/active flags.	Branch is available within assigned divisions.
F-ORG-04	Configure branch-division	Local contacts, manager, services, booking acceptance, maintenance capability, workflow, terms and price overrides.	Valid operating combination is created.
F-ORG-05	Operating calendar	Weekly hours, pickup/return cutoff, holidays, temporary closures and after-hours charge.	Availability and pricing respect calendar.
F-ORG-06	Future division enablement	Create division → services → categories/attributes → branches → rates/workflows → staff scopes.	New hire line can operate without changing core rental tables.

6.1 Initial service configuration

Division	Example services	Default charging / special rules
Motors	Vehicle rental	Daily/weekly; driver licence and odometer/fuel inspections.
Carptrac	Forklift, crane, excavator, generator hire	Hourly/daily; meter readings; qualified operator optional/required by service.
Shipping limited hire	Portable toilet, large bin and scaffolding hire	Day/week/month/unit; delivery/collection, cleaning, setup, labour and quantity/site details.

7. Customer records and customer portal

Maintain one authoritative customer record while separating rental eligibility from online account security.

Specification item	Definition
Primary actors	Customer, Rental Officer, Administrator, Finance Officer
Required permissions / scope	customers.manage_access for portal security; operational and financial permissions by function and scope.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

ID	Function	Processing / rules	Outcome
F-CUS-01	Create/search customer	Search before create; unique customer number; classify Individual or Corporate; capture contacts, IDs/business details, address and status.	Customer profile created without duplicate.
F-CUS-02	Corporate account	Multiple contacts, approved renters, PO requirement, payment terms, credit limit, contracted rates and consolidated activity.	Corporate controls applied to bookings.
F-CUS-03	Enable portal access	Invite existing customer or register and link after verification; portal state independent from business/rental status.	Activation email and auditable account link.
F-CUS-04	Customer sign-in	Customer credential flow is separate from /staff; only own/corporate-authorized data is returned.	Customer portal session.
F-CUS-05	Portal workspace	Show bookings, quotation, agreement, invoice/payment status, documents and activity; allow cancellation/extension/incident requests.	Request is recorded for staff decision.
F-CUS-06	Security actions	Resend activation, reset password, verify email, lock/unlock portal, revoke sessions and view failed attempts.	Security state changes without changing rental eligibility.
F-CUS-07	Public status lookup	Allow reference-based status lookup only with a second matching value or signed secure link.	Minimal non-sensitive status returned; no customer PII exposed.

7.1 Customer field rules

Field group	Individual	Corporate	Validation
Identity	Full name, DOB where approved, ID type/number, driver licence when driving.	Legal name, registration/TIN where approved, account contacts and authorized renters.	Required fields vary by service; sensitive values masked in lists.
Contact	Email, mobile, address.	Billing and operational contacts, email, phone and addresses.	Email format; phone normalized; verified flags stored separately.
Commercial	Deposit/payment status.	Credit limit, available credit, terms, PO rule and contract rate.	Credit/eligibility checked before confirmation.
Access	Portal enabled, email verified, account state.	Contact-level portal authority.	Rental status and portal status remain independent.

8. Asset register, QR, lifecycle and availability

Maintain a complete operational and financial record for each rentable asset across all configured divisions.

Specification item	Definition
Primary actors	Rental Officer, Branch Manager, Maintenance Officer, Driver/Operator, Finance and authorized administrators
Required permissions / scope	assets.view/create/edit/transfer/inspect/record_meter/retire/view_financials as applicable.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

ID	Function	Processing / validation	Result
F-AST-01	Create asset	Unique asset number; division, branch, service and category required; capture registration/serial/VIN/engine, make/model/year, ownership, acquisition, insurance, warranty, location and image/doc references.	Commissioned asset profile.
F-AST-02	Dynamic attributes	Render category definitions by type: text, number, integer, date, yes/no or choice; required/unit/search/report/customer visibility flags.	Future asset types need configuration rather than schema changes.
F-AST-03	QR identification	Generate a non-sensitive QR code resolving to asset number/token; scan checks authorization and returns allowed next actions.	Fast verified asset selection.
F-AST-04	Meter reading	Record odometer, engine hours, operating hours, fuel or units with source, date and actor; reject rollback unless authorized correction reason.	Current reading and history updated.
F-AST-05	Lifecycle transition	Validate allowed transition and prerequisites; capture actor, timestamp, booking, meter, fuel, notes and evidence.	Status, lifecycle and audit updated atomically.
F-AST-06	Transfer / retirement	Require origin/destination, dispatch/receipt checks; retirement requires reason, approval and final values.	Availability and location reflect transfer/retirement.
F-AST-07	Performance profile	Display revenue, service/transport/operator revenue, costs, margin, utilization, downtime, cost per km/hour and break-even progress.	Financial tab shown only with permission.

8.1 Availability engine

Conflict source	Blocking condition	User-facing response
Booking	Overlapping non-cancelled/non-expired booking item.	Unavailable with booking reference to authorized staff.
Maintenance	Open/in-progress/waiting job overlapping requested period.	Unavailable — maintenance; expected release if known.
Inspection	Required inspection incomplete or asset status Inspection.	Unavailable until checklist passes.
Transfer	Asset in transit or assigned to another branch for requested period.	Unavailable at selected branch; show expected location.
Branch calendar	Closure/cutoff incompatible with pickup/return.	Offer valid time/branch; apply after-hours rule if allowed.
Personnel	Required operator unavailable or qualification invalid.	Do not confirm until eligible person is allocated.

9. Booking requests, quotations and approvals

Convert customer demand into a correctly priced, conflict-free and approved booking using a visible work queue.

Specification item	Definition
Primary actors	Customer, Rental Officer, Branch Manager, authorized approvers
Required permissions / scope	Manage rentals within scope; rentals.approve and pricing.configure where applicable.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

9.1 Request and work-queue flow

ID	Step	Required inputs / checks	Next state / action
F-BKG-01	Submit request	Customer, division/service, asset/category, branch, start/end, purpose/site and required personnel/delivery details.	New request; reference issued and acknowledgement queued.
F-BKG-02	Triage request	Check duplicates, customer eligibility, credit/PO flags, service documents and preliminary availability.	Prepare quotation, request information or reject with reason.
F-BKG-03	Allocate asset	Select available in-scope asset; detect booking, maintenance, transfer, inspection and calendar conflict.	Asset reserved provisionally or alternatives shown.
F-BKG-04	Confirm booking	Accepted quote, approval complete, customer eligible and no current conflict.	Confirmed; pickup requirements generated.
F-BKG-05	Cancel/expire	Record permitted reason; release reservations and notify affected parties.	Cancelled or expired; audit and availability updated.

9.2 Pricing and quotation engine

Calculation stage	Rule
Rate resolution	Corporate contract → branch override → division rate → category/asset rate → standard default. Most specific active rate valid for the hire period wins.
Hire duration	Calculate billable quantity using configured unit and minimum hire period; respect partial-day/hour and branch calendar rules.
Adjustments	Apply weekend, holiday and overtime multiplier; included usage and excess km/hour; operator/driver, delivery zone/distance/trip, setup, labour, fuel and other configured charges.
Commercial controls	Apply approved discount, deposit and customer terms; distinguish customer-visible selling rate from internal cost rate.
Tax	Apply configured tax rate to taxable lines; support tax-inclusive or tax-exclusive presentation. Rounding occurs per approved finance rule.
Quote total	Subtotal – discount + additional charges + tax. Deposit is displayed separately unless policy includes it in amount due.
Revision	Every revision receives version, creator, timestamp, reason and line snapshot; issued revisions remain read-only.
Conversion	Only accepted, unexpired, approved quotation converts; conversion is idempotent and creates/updates the booking once.

9.3 Configurable approval workflow

ID	Function	Business behaviour
F-APR-01	Workflow definition	Configure division/branch, trigger, value band, number/order of stages, approver role or named employee, SLA and active dates.
F-APR-02	Submission	Create immutable approval instance from the applicable workflow; freeze decision inputs and assign stage 1.
F-APR-03	Decision	Approver may approve, reject or request revision with reason. Actor must be eligible, active and not prohibited by separation-of-duties rules.
F-APR-04	Progression	Approval advances sequentially; booking/quote remains pending until all required stages approve.
F-APR-05	Delegation	Use an active dated delegate when the primary approver is unavailable; record both the original assignment and acting approver.
F-APR-06	Escalation	Scheduled job flags or reassigns delayed stages according to policy and notifies management.
F-APR-07	Optional workflow	Where no applicable workflow exists, authorized low-risk transactions can proceed without approval; reason is recorded.

9.4 Booking workspace layout

Panel	Content	Primary action examples
Summary	Reference, customer, division/service, branch, dates, value, stage and warnings.	Prepare quotation / Submit approval / Confirm.
Customer	Contacts, status, eligibility, credit and required documents.	Resolve missing information.
Asset & availability	Requested category, allocated asset, conflict explanation and alternatives.	Allocate or replace asset.
Pricing	Customer-visible lines, deposit, discount, tax, total; internal cost limited by permission.	Revise or issue quotation.
Approval & documents	Stage history, decision reasons, quote versions, attachments.	Submit / decide / send.
Activity	Notes, emails, status and audit timeline, collapsed by default.	Add internal note.

10. Pickup, agreements, active hire and returns

Use guided, controlled workflows to place an asset on hire and return it with complete evidence and correct financial consequences.

Specification item	Definition
Primary actors	Rental Officer, Driver/Operator, Customer, Branch Manager, Finance and Maintenance staff
Required permissions / scope	Rental-operation access plus asset inspection/meter and approval permissions where needed.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

10.1 Pickup and signed agreement

Step	System requirement	Completion result
1 Confirm customer	Booking confirmed; customer identity, authorized renter and contact checked.	Customer prerequisite complete.
2 Confirm/scan asset	QR resolves to allocated asset and branch; no conflict or blocking status.	Correct asset locked to workflow.
3 Verify documents	Licence/ID and service documents verified; corporate PO/credit requirements satisfied.	Compliance prerequisite complete.
4 Record condition	Pre-hire template complete with meter, fuel, responses, evidence and damage baseline.	Passed inspection or maintenance/damage referral.
5 Review charges	Accepted quote, deposit/payment requirement and approved changes displayed.	Agreement snapshot ready.
6 Sign and approve	Customer signs; agent verifies and approves. Signature consent, signer name and timestamp retained.	Agreement becomes immutable.
7 Check out	Generate PDF, queue email, write lifecycle/audit, status On hire/Rented.	Active hire appears in work queue.

Agreement integrity: A signed agreement is never edited. Corrections use a numbered addendum linked to the original, with reason, approver and signatures where required.

10.2 Active hire queues

Queue	Included records	Displayed next action
Pickup today	Confirmed hires due for collection, incomplete prerequisites visible.	Continue pickup.
On hire	Checked-out assets not yet due.	View hire / report incident / extension request.
Due today	Expected return today.	Start return.
Overdue	Return date/time passed and not completed.	Contact customer / calculate late charge / escalate.
Return in progress	Post-hire workflow started but not finalized.	Continue return.
Recently completed	Completed within configured period.	View invoice, documents and history.

10.3 Return workflow

Step	Processing and validation	System output
1 Scan and identify	QR must match active hire asset; otherwise require authorized exception.	Correct booking/customer loaded.
2 Record return	Capture actual date/time, location, meter and fuel; readings cannot be below handover without correction reason.	Usage and lateness calculated.
3 Post-hire inspection	Complete category checklist; compare baseline; capture photos, damage, customer acknowledgement and notes.	Outcome Passed / Passed with notes / Failed / Damage detected.
4 Calculate adjustments	Late time, excess usage, fuel, cleaning, damage and other approved lines.	Final charge preview with internal/customer separation.
5 Refer issues	Fault/damage can create linked maintenance job or damage incident.	Asset placed in Inspection, Maintenance or Out of service.
6 Complete return	Finalize invoice; save signatures; create lifecycle and audit records.	Booking Completed; asset status determined automatically.
7 Customer output	Generate/send return summary and invoice or amount-due notice.	Delivery event visible to staff.

10.4 Automatic next-status rule

Condition after return	Next asset status
Inspection passed and no maintenance due	Available
Minor notes require review	Inspection
Fault, failed checklist or maintenance threshold reached	Maintenance
Unsafe or serious damage	Out of service
Approved retirement/disposal workflow	Retired

11. Inspections and maintenance

Provide configurable evidence-based checks and detailed maintenance expenditure so asset condition, downtime and profitability remain trustworthy.

Specification item	Definition
Primary actors	Rental Officer, Driver/Operator, Maintenance Officer and Branch Manager
Required permissions / scope	assets.inspect, assets.record_meter, maintenance.complete and asset-scope access.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

11.1 Inspection templates

Asset group	Checklist examples	Mandatory evidence / special handling
Motors vehicle	Exterior/body, tyres, glass/lights, interior, documents, tools, fuel, odometer and existing damage map.	Pre/post signatures; licence and ID at pickup; photos for damage.
Carptrac forklift/crane/excavator	Fluids/leaks, tyres/tracks, forks/boom/attachments, hydraulics, brakes, alarms, safety devices and engine hours.	Qualified operator check; failed safety item blocks hire.
Carptrac generator	Oil/coolant/fuel, battery, cables, controls, output, voltage/phase, operating hours and test run.	Reading and test result; failed electrical/safety item blocks hire.
Shipping portable toilet	Shell/door/lock, tank, cleanliness, consumables, site condition and unit quantity.	Delivery/collection photos; cleaning/service requirement.
Shipping large bin	Structure, floor, doors/locks, contamination, damage, contents restriction and site access.	Photos, quantity, delivery/collection condition.
Shipping scaffolding	Component quantities, frames/braces/planks, damage/corrosion, load/safety tags and returned count.	Quantity reconciliation; missing/damaged components become charge/referral.

11.2 Maintenance processing

ID	Function	Rules and outputs
F-MNT-01	Create maintenance job	Asset, job number, type, priority, issue, reported date and source required; set asset Maintenance/Out of service when blocking.
F-MNT-02	Plan preventive work	Rules may be every N kilometres, N engine hours, N months or whichever occurs first; scheduled job created at threshold.
F-MNT-03	Record work and cost	Labour hours/rate, parts and quantity/cost, supplier, external invoice, transport/fuel/other cost, warranty and notes.
F-MNT-04	Manage status	Open → In progress → Waiting for parts → Completed/Cancelled. Completion requires work summary and authorization.
F-MNT-05	Return to service	Safety/inspection requirement must pass; meter and next-service targets saved; asset returns Available only if no other blocker.
F-MNT-06	Analyze history	Show repeat failures, downtime, total cost, maintenance cost versus replacement/book value and upcoming work.

12. Personnel, transport and field operations

Coordinate operators, drivers and delivery activity required by equipment and site-based hire services.

Specification item	Definition
Primary actors	Driver, Operator, Rental Officer, Branch Manager and authorized administrators
Required permissions / scope	Scoped assignment access; personnel configuration and timesheet privileges.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

ID	Function	Functional rules
F-OPS-01	Personnel profile	Store employee/reference, branch/division, contact, hourly/overtime cost, availability and active state.
F-OPS-02	Qualifications	Record licence/certificate type, number, issue/expiry and evidence; expired/incorrect qualification blocks assignment.
F-OPS-03	Assign operator/driver	Check overlap, availability, qualification, branch/division and booking period; retain role and planned hours.
F-OPS-04	Timesheet	Assigned person records start/end/break/overtime and notes; manager approval determines payable/chargeable hours.
F-OPS-05	Delivery/collection	Capture zone, address/site contact, truck/driver, scheduled window, distance/trips and instructions; status Planned → Dispatched → Delivered/Collected/Failed.
F-OPS-06	Proof of service	Capture QR, photos, meter/unit count, recipient name/signature, timestamp and failure reason.
F-OPS-07	Cost and revenue	Separate customer operator/transport charge from employee/vehicle/internal cost for profitability.

12.1 Mobile field workspace

- Show only the signed-in user's current and upcoming assignments.
- Allow QR resolve, checklist, meter, time, photograph and signature capture on responsive screens.
- Do not expose customer financials, unrelated assets or other personnel records.
- When connectivity fails, preserve unsent form state locally only if an approved secure offline design is implemented; baseline does not claim offline synchronization.

13. Finance, profitability, dashboards and reports

Produce reliable customer charges and management insight while keeping internal costs and financial actions permission-controlled.

Specification item	Definition
Primary actors	Finance Officer, Branch Manager, Division/Group Management and authorized administrators
Required permissions / scope	reports.financial, payments.refund, assets.view_financials and scoped operational access.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

13.1 Invoices, payments and statements

ID	Function	Rules / result
F-FIN-01	Invoice line editor	Draft/issued unpaid invoice lines include description, quantity, unit price, tax rate and taxable flag; Paid/Voided invoices are locked.
F-FIN-02	Totals	Subtotal=sum quantity×unit price; tax follows inclusive/exclusive setting; total and balance update after every permitted change.
F-FIN-03	Payment allocation	Positive allocation cannot exceed unallocated payment or invoice balance; update Partially paid/Paid atomically.
F-FIN-04	Credit note/refund	Require reason, permission and approval where configured; value cannot exceed eligible invoice/payment amount.
F-FIN-05	Credit control	Before confirmation, compare exposure against limit/terms and PO requirement; route exception to approval.
F-FIN-06	Statement	Show customer invoices, credits, allocations, total, paid and outstanding across authorized divisions; provide PDF/email output.
F-FIN-07	Accounting export	Export approved transaction fields with stable IDs, dates, tax, branch/division and reference for later accounting integration.

13.2 Asset profitability

Metric	Calculation / source
Rental and service revenue	Customer-visible earned hire, operator, delivery, excess usage, cleaning/damage recovery and other approved revenue attributed to asset.
Operating cost	Maintenance parts/labour/contractor, operator, transport, fuel, cleaning, insurance, registration, storage, damage and configured allocations.
Gross profit	Attributed revenue – attributed operating cost.
Profit margin	Gross profit ÷ attributed revenue × 100; blank/NA when revenue is zero.
Cost per unit	Relevant operating cost ÷ valid km/hour/day/unit usage; never divide by zero.
Utilization	Billable on-hire time or units ÷ available rentable capacity for the selected period.
Trend / warning	Monthly series; flag loss-making assets using configurable threshold and period.

13.3 Role-specific dashboards and reports

Audience	Dashboard priorities	Reports / outputs
Rental Officer	Today's requests, pickups, returns, incomplete prerequisites and overdue hires.	Rental history, availability and customer rental history.
Branch Manager	Approvals, conflicts, overdue, low availability, utilization and branch revenue/expense/profit.	Branch fleet/equipment utilization, overdue and profitability.
Maintenance Officer	Due/overdue service, faults, waiting parts and downtime.	Maintenance expenditure, repeat failure and downtime.
Finance Officer	Invoices due, unallocated payments, credit issues and exceptions.	Revenue, outstanding balances, customer profitability and statements.
Administrator	Configuration/access exceptions, email failures, audit and system health.	Access, audit and configuration reports.
Management	Division/branch performance, assets producing losses and trends.	Revenue by division/branch/category; asset P&L; utilization; export CSV/Excel/PDF.
Customer	Own upcoming/active/past bookings, quote/agreement/invoice and requests.	Own documents and account statement where enabled.

13.4 Report controls

- Date range, division, branch, service, category, asset and customer filters are constrained by actor scope.
- Report totals use the same source records and calculation services as operational screens.
- Exports carry report title, generation time, filters, currency/tax basis and user scope.
- Large exports run asynchronously or stream results; screen reports use server aggregation and pagination.

14. Notifications, documents and scheduled processing

Deliver required communications reliably, generate consistent records and automate time-based operational controls.

Specification item	Definition
Primary actors	Customers, staff, administrators and system scheduler
Required permissions / scope	Template/configuration permissions; recipient and record scope checked before sending.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

Event	Recipient / channel	Content and rule
Account activation / reset / verification	Target email	Single-use expiring link/code; generic public response; never include password.
Booking request / confirmation	Customer and relevant branch staff	Reference, service, dates, branch, next step and secure portal link.
Quotation	Authorized customer contacts	Versioned PDF/secure link, expiry and acceptance instructions.
Agreement / addendum	Customer and branch records	Immutable signed PDF; addendum separately numbered.
Invoice / statement	Authorized billing contacts	PDF/secure download and due/balance information.
Operational alerts	Scoped staff/manager	Overdue, maintenance due, low availability, approval delay, expiry, unpaid invoice, excess usage or unresolved damage.
Delivery failure	Administrator/operations	Failure category, retry state and safe diagnostic reference; no secret values.

14.1 Scheduled jobs

Job	Frequency / idempotency	Result
Expire quotations	Configured interval; repeat does not duplicate state/event.	Open expired quotes marked Expired; reservations released if policy says so.
Detect overdue rentals	Frequent interval based on branch local time.	Overdue flag/alert and notification queue.
Maintenance threshold	Daily and after meter entry.	Due/overdue job or reminder created once.
Credential/document expiry	Daily look-ahead.	Licence, insurance and warranty alerts.
Delayed approvals	Configured SLA interval.	Escalation/delegation and notification.
Email delivery	Queue polling with bounded retries.	Sent, failed or retry state plus provider reference.
Availability/profit flags	Scheduled management refresh.	Low fleet and loss-making alerts without changing source financial data.

14.2 Document control

- Store document metadata and private storage reference; do not store large data URLs in operational tables for production.
- Validate file type, size and content; use private signed download links and access auditing.
- Generated PDF stores template version and source-record snapshot/hash.
- Retention, malware scanning and versioning require production policy confirmation.

15. API, validation, errors and audit behaviour

Define consistent service behaviour so web clients, tests and future integrations interact safely and predictably.

Specification item	Definition
Primary actors	Web clients, authorized integrations, administrators and support staff
Required permissions / scope	Endpoint policy plus server-side permission and scope validation.
Entry condition	Authenticated actor has an active account and access to the selected division/branch unless the function is public.
Common exit condition	Data is validated, saved atomically, audit history is recorded and the next permitted action is shown.

15.1 API conventions

Concern	Required behaviour
Transport	HTTPS in staging/production; JSON UTF-8; /api route prefix; explicit request/response contracts.
Authentication	Secure, HttpOnly, SameSite cookie plus window-bound session token where designed; CSRF protection for cookie-authenticated mutations.
Timeouts	Inbound request timeout and downstream email/document/database timeouts are bounded. Long reports/jobs use asynchronous processing rather than keeping requests open.
Validation	Reject malformed/invalid input before mutation; trim text; validate ranges, dates, ownership, state and concurrency on server.
Pagination	List endpoints accept page/pageSize/search/filter/sort within configured limits and return total/page metadata.
Idempotency	Quote conversion, payment allocation, checkout/return and external retry-sensitive actions resist duplicate submission.
Concurrency	Use version/concurrency token or re-read status for sensitive transitions; stale updates return conflict.
Data exposure	Return only required fields; financial/internal cost and cross-scope records excluded at query level.
Correlation	Every request receives correlation ID propagated to safe errors, audit and logs.
Rate limiting	Login, password reset, public booking/status and costly export endpoints use stricter limits.

15.2 Error contract

HTTP status	Meaning	Client behaviour
400	Validation/business input error	Show field/reason; preserve user input.
401	Missing, expired or invalid session	Clear local session and return to correct login.
403	Authenticated but not permitted/in scope	Explain access restriction without exposing record data.
404	Record not found or deliberately concealed across scope	Show not found; do not infer existence.
409	State/concurrency/availability conflict	Refresh workspace and show current state or alternatives.
422	Optional for multi-rule validation	Show business-rule failures as structured issues.
429	Rate limit exceeded	Show retry guidance; do not loop automatically.
500/503	Unexpected/unavailable dependency	Safe message, correlation ID and bounded retry option.

15.3 Audit and logging

Event category	Minimum audit fields	Additional rule
Security	Actor/target, outcome, time, IP/device, session/challenge reference.	Passwords, OTPs, reset tokens and secrets are never logged.
Access/configuration	Actor, target, before/after, role/permission/scope, reason.	Super Administrator and division changes are high priority.
Rental lifecycle	Booking/asset/customer references, action, old/new status, time and branch/division.	Signed agreement and inspection versions linked.
Financial	Invoice/payment/credit reference, amount/currency, actor, approval and before/after balance.	Append-only business audit; correction is a new event.
Maintenance/inspection	Asset/job/template, outcome, meter, cost, evidence references and actor.	Evidence access is separately audited.
System/integration	Job, correlation ID, attempt, provider status, duration and error category.	Production logs exclude personal message bodies unless approved.

15.4 Transaction boundaries

- Checkout saves inspection/agreement state, booking status, asset status, lifecycle and audit together; failure rolls back business-state changes.
- Return saves inspection, final charges/invoice state, booking/asset status, maintenance referral and audit consistently.
- Email delivery is queued after the business transaction; email failure does not undo a valid checkout or payment.
- Approval decisions and payment allocations use database constraints/concurrency checks to prevent duplicates.

16. Functional acceptance and traceability

16.1 Acceptance scenarios

Scenario	Given / when	Expected observable result
AC-01 Scoped staff login	Branch Manager assigned to Motors Suva signs in.	Only permitted Motors/Suva work, assets, reports and menu items are returned; administration branches page is not accessible.
AC-02 Customer booking	Verified customer selects division/service/dates and submits complete request.	Reference issued, request visible in correct branch queue and confirmation email queued.
AC-03 Conflict prevention	Officer allocates asset overlapping maintenance or confirmed booking.	409/conflict shown with reason; booking is not confirmed.
AC-04 Flexible equipment pricing	Carptrac forklift requires operator and delivery.	Quote includes hire, operator hours and transport with customer total and separate internal cost.
AC-05 Multi-stage approval	High-value quote matches three-stage workflow.	Stages process sequentially; creator cannot self-approve; confirmation waits for final approval.
AC-06 Vehicle checkout	Confirmed Motors booking has verified ID/licence, payment and passed pre-hire check.	Signed immutable PDF generated, email queued, asset On hire and lifecycle/audit recorded.
AC-07 Damaged return	Post-hire comparison records new damage.	Damage/maintenance referral created, final charges reviewed, asset not Available.
AC-08 Preventive maintenance	New meter reaches configured threshold.	One maintenance reminder/job is created and shown to scoped maintenance staff.
AC-09 Shipping hire	Portable toilet order includes quantity, delivery and cleaning.	Availability/quantity validated; quote includes unit hire, delivery and cleaning; delivery proof supported.
AC-10 Asset profitability	Manager filters one asset and month.	Revenue, attributable costs, profit, margin and utilization reconcile to underlying records.
AC-11 Session security	Idle timeout passes or session cookie copied to unmatched window.	Protected request receives 401 and no protected data is shown.
AC-12 Failed email	Provider rejects quotation email.	Quote remains issued; email marked failed/retryable; staff sees delivery state and audit reference.

16.2 Functional traceability matrix

Business objective	FSD functions	Evidence for acceptance
Secure access and isolation	F-AUTH-01-05, F-IAM-01-06	Authorization tests, session tests, scope-isolation tests and audit review.
Flexible multi-division operation	F-ORG-01-06, F-AST-02	Configured Motors, Carptrac and limited Shipping services without core schema changes.
Complete asset tracking	F-AST-01-07, inspection and return functions	Asset profile, QR, meters, inspections, lifecycle and status history.
Reliable hire workflow	F-BKG-01-05, F-APR-01-07, Sections 10-12	End-to-end request-to-return test with documents and assignments.
Financial visibility	F-FIN-01-07 and profitability metrics	Reconciled invoice/payment/asset P&L examples and exports.
Operational control	Sections 11, 13 and 14	Maintenance rules, dashboards, scheduled alerts and report filters.
Accountability	Section 15.3	Audit records for security, configuration, lifecycle and finance actions.

16.3 Definition of functionally complete

- All mandatory flows and negative cases pass in a production-like staging environment.
- Permissions and division/branch isolation are verified with representative accounts.
- Carpenters confirms field names, forms, checklists, statuses, price/tax rules, documents and reports.
- Email/PDF/QR behaviours are tested with approved non-production credentials and recipients.
- No critical/high defects remain; accepted lower-priority limitations are documented.

17. Assumptions, open decisions and change control

17.1 Assumptions

ID	Assumption	Impact if incorrect
AS-01	FJD and 15% tax are defaults but configurable.	Pricing and document totals require revision.
AS-02	A branch may support multiple divisions; services are enabled per branch.	Organization/scoping model changes.
AS-03	Shipping scope is limited to portable toilets, large bins and scaffolding hire.	Additional shipping processes require change request.
AS-04	Signed agreements occur at pickup after prerequisites pass.	Agreement lifecycle and UI changes.
AS-05	Email is a notification channel; business transactions remain valid if delivery fails.	Transaction/integration design changes.
AS-06	Customer online access and rental eligibility are separate statuses.	Customer lifecycle and controls change.

17.2 Decisions required before final sign-off

Decision	Carpenters confirmation required
Pricing/tax rounding	Line versus invoice rounding, tax-inclusive/exclusive defaults, partial-day rules and deposit treatment.
Approval thresholds	Value bands, approver roles/names, stages, SLA, escalation and delegation.
Customer verification	Approved identity/business fields, retention, masking and public tracking verification.
Inspection templates	Final checklist items, mandatory photos/signatures, failure rules and damage recovery.
Shipping operations	Quantity inventory model, delivery zones, cleaning/setup rules and component reconciliation.
Security/session	Idle/absolute timeout, MFA policy, window-bound behaviour, password policy and lockout thresholds.
Documents/email	Approved templates, sender domain, recipient rules, retention and legal wording.
Finance integration	Source of truth, invoice numbering, payment process and export format.
Data migration	Validated mapping, cleansing, ownership and import acceptance criteria.

17.3 Change-control workflow

- Log change with rationale, requester, affected module and urgency.
- Assess functional, security, data, UI, test, schedule and cost impact.
- Obtain authorized business and project approval before implementation.
- Update FSD/SRS version, traceability and acceptance cases; communicate the baseline.

18. Client sign-off

The undersigned confirm that this Functional Specification Document accurately describes the intended CREMS behaviour for the agreed baseline, subject to documented comments and approved change control.

Approval statement	Selection / comment
Functional scope and workflows	☐ Approved ☐ Approved with changes ☐ Not approved
Roles, permissions and data visibility	☐ Approved ☐ Approved with changes ☐ Not approved
Pricing, approvals and finance behaviour	☐ Approved ☐ Approved with changes ☐ Not approved
Inspection, maintenance and asset lifecycle	☐ Approved ☐ Approved with changes ☐ Not approved
Documents, email and security behaviour	☐ Approved ☐ Approved with changes ☐ Not approved

Name	Role	Signature	Date

Sign-off comments / required amendments

Appendix A — Status models

Object	Primary status sequence	Exceptions / notes
Staff/customer account	Invited → Activation pending → Active → Locked / Suspended / Deactivated	Reactivation requires authorized action; departure revokes sessions.
Booking	Draft / New request → Quotation/approval activity → Confirmed → Converted to rental → Completed	Cancelled and Expired are terminal; revisions remain in history.
Asset	Commissioned → Available → Reserved → Inspection → Rented/On hire → Inspection → Available	May branch to Maintenance, Out of service, transfer or Retired.
Maintenance	Open → In progress → Waiting for parts → Completed	Cancelled requires reason; return-to-service is separate validation.
Invoice	Draft → Issued → Partially paid → Paid	Voided/credit corrected by controlled finance actions.
Approval	Pending → Approved / Rejected / Cancelled	Multi-stage instance completes only after all required approvals.
Delivery	Planned → Dispatched → Delivered / Collected / Failed	Proof and failure reason required according to service.

Appendix B — Output document catalogue

Document	Generation trigger	Control
Quotation PDF	Issue or approved revision	Versioned, expiry shown, accepted version linked to booking.
Booking confirmation	Booking confirmation	Reference, service, dates, branch and prerequisites.
Rental agreement PDF	Customer signature and agent approval at pickup	Immutable snapshot; corrections by addendum.
Inspection report	Checklist completion	Template version, responses, readings, signatures and evidence references.
Return summary	Return completion	Usage, condition differences, adjustments and next asset status.
Invoice / credit note	Finance issue/action	Unique number, tax basis, lines, totals and balance.
Customer statement	Requested/scheduled corporate statement	Period, invoices, credits, payments and outstanding.
QR label	Asset commissioning/reprint	Non-sensitive token/code and human-readable asset number.
Management export	Authorized report export	Filters, scope, currency, generated time and user.

Appendix C — Functional requirement index

Range	Module	Principal functions
F-AUTH-01-05	Authentication	Login, MFA, window-bound session, timeout and logout/revocation.
F-IAM-01-06	Identity and access	Staff lifecycle, scopes, permissions, security actions and separation of duties.
F-ORG-01-06	Organization	Divisions, services, branches, branch configuration, calendar and expansion.
F-CUS-01-07	Customers	Customer/corporate records, portal, access security and status tracking.
F-AST-01-07	Assets	Register, dynamic attributes, QR, meters, lifecycle, transfers and financial profile.
F-BKG-01-05	Bookings	Request, triage, allocation, confirmation and cancellation/expiry.
F-APR-01-07	Approvals	Workflow definition, submission, decisions, progression, delegation and escalation.
F-MNT-01-06	Maintenance	Jobs, preventive rules, work/cost, status, return to service and history.
F-OPS-01-07	Personnel/logistics	Profiles, qualifications, assignments, timesheets, delivery and proof.
F-FIN-01-07	Finance	Invoice, totals, allocation, credit/refund, controls, statements and export.

End of controlled document: Any approved amendment must update the version history, affected function IDs and acceptance evidence.